Quantitative Comparisons

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

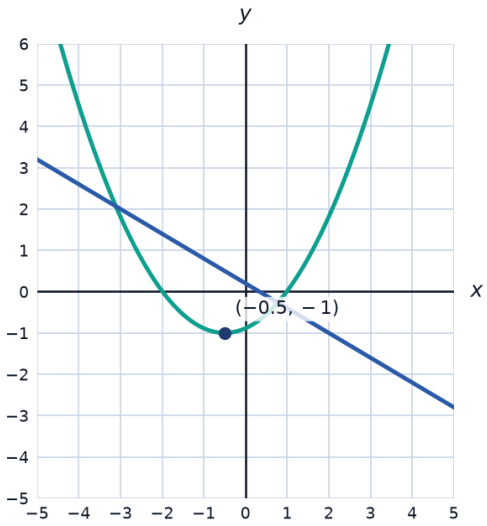
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Numeric comparisons

Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

2 [Foundational] • Target: Numeric comparisons

Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.



3 [Foundational] • Target: Numeric comparisons

Column A: 9^2. Column B: $8^2 + 15$. Compare the two columns.

4 [Foundational] • Target: Numeric comparisons

Column A: $(-3)^2$. Column B: -3^2. Compare the two columns.

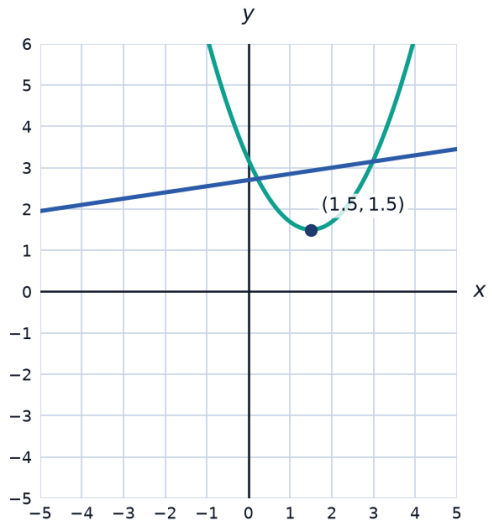
5 [Foundational] • Target: Fraction and decimal comparisons

Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

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6 [Foundational] • Target: Fraction and decimal comparisons

Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.



7 [Foundational] • Target: Fraction and decimal comparisons

Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Percent comparisons

Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

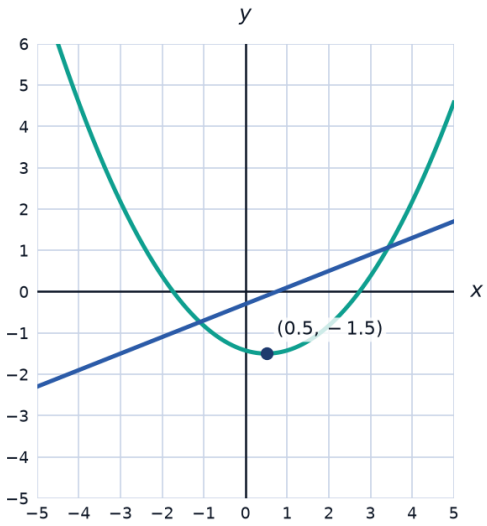
9 [Intermediate] • Target: Percent comparisons

Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

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10 [Intermediate] • Target: Exponent and root comparisons

Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.



11 [Intermediate] • Target: Exponent and root comparisons

Column A: 2^{10}. Column B: 10^3. Compare the two columns.

12 [Intermediate] • Target: Exponent and root comparisons

Column A: 3^4. Column B: 4^3. Compare the two columns.

13 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.

14 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

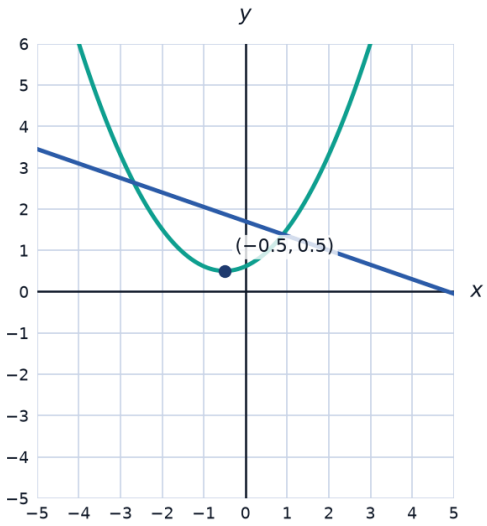
15 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 6.01 Sequences & Patterns

Given $n = -2$. Column A: n^2. Column B: n^3. Compare the two columns.

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16 [Challenging] • Target: Algebraic comparisons at a fixed value • Cross-Lesson: 7.01 Data Analysis & Probability

Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.



17 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 8.01 Word Problems & Reasoning

For any real number x: Column A: x. Column B: x^2. Compare the two columns.

18 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 1.01 Arithmetic & Number Sense

For any real number x: Column A: $x + 5$. Column B: $2x$. Compare the two columns.

19 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

For any real number x: Column A: $|x|$. Column B: x. Compare the two columns.

20 [Challenging] • Target: Comparisons requiring testing values • Cross-Lesson: 3.01 Algebra Essentials

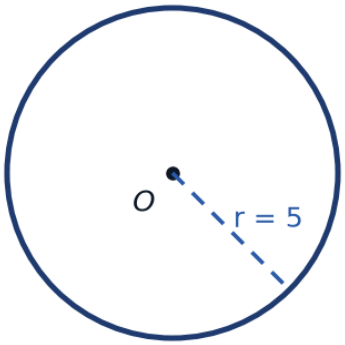
Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Geometry comparisons

Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.



a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater

22 [Foundational] • Target: Geometry comparisons

Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

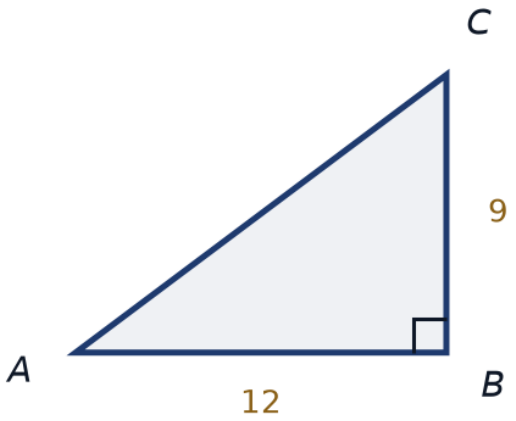
c. Column B is greater

d. Column A is greater

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23 [Foundational] • Target: Geometry comparisons

Column A: the hypotenuse of the right triangle shown. Column B: 16.



a. The relationship cannot be determined from the given information

b. Column B is greater

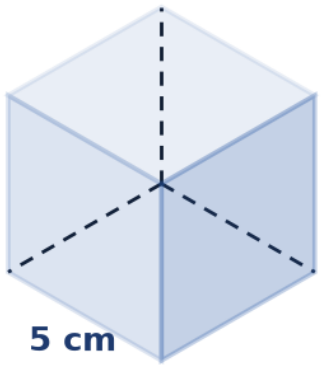
c. The two columns are equal

d. Column A is greater

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24 [Foundational] • Target: More geometry comparisons

Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.



a. The relationship cannot be determined from the given information

b. Column A is greater

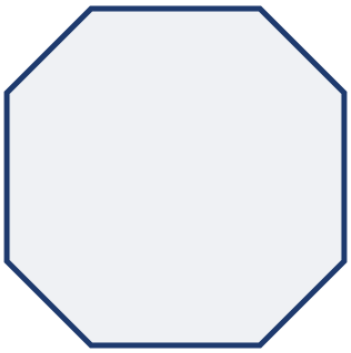
c. Column B is greater

d. The two columns are equal

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25 [Foundational] • Target: More geometry comparisons

Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.



a. Column B is greater

b. The two columns are equal

c. Column A is greater

d. The relationship cannot be determined from the given information

26 [Foundational] • Target: Numeric comparisons

Column A: 6×7. Column B: $5 \times 9 - 3$. Compare the two columns.

a. The two columns are equal

b. Column B is greater

c. The relationship cannot be determined from the given information

d. Column A is greater

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27 [Foundational] • Target: Numeric comparisons

Column A: $345 + 678$. Column B: $900 + 100$. Compare the two columns.

a. Column A is greater

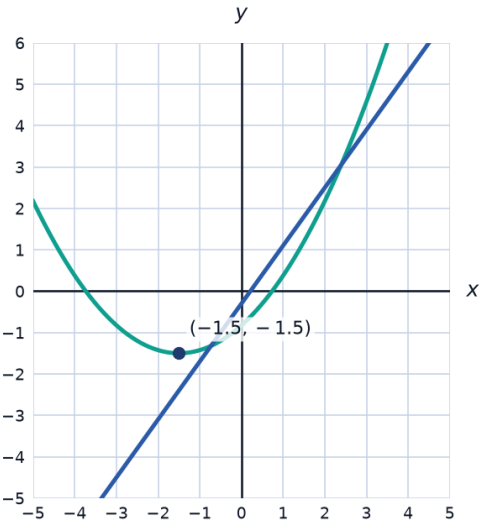
b. The relationship cannot be determined from the given information

c. Column B is greater

d. The two columns are equal

28 [Foundational] • Target: Percent comparisons

Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.



a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. Column B is greater

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 6$. Column A: $3x - 4$. Column B: $x + 10$. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The relationship cannot be determined from the given information

d. The two columns are equal

30 [Intermediate] • Target: Algebraic comparisons at a fixed value

Given $x = 2$. Column A: $4x + 1$. Column B: $3(x + 2)$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater

31 [Intermediate] • Target: Comparisons requiring testing values

For any real number x: Column A: x^2. Column B: 0. Compare the two columns.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. Column B is greater

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32 [Intermediate] • Target: Comparisons requiring testing values

For any integer n: Column A: n. Column B: $-n$. Compare the two columns.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column B is greater

d. Column A is greater

33 [Intermediate] • Target: Comparisons requiring testing values

For any real number x with $x \neq 0$: Column A: x^2. Column B: 0. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information

34 [Intermediate] • Target: Comparisons requiring testing values

For any real number x: Column A: $(x+2)^2$. Column B: x^2+4x+4. Compare the two columns.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

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35 **[Intermediate]** • **Target: Comparisons requiring testing values**

Given only that m is a positive integer: Column A: m. Column B: m^2. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information

36 **[Intermediate]** • **Target: Comparisons requiring testing values**

For any real number x: Column A: $-x$. Column B: $x - 1$. Compare the two columns.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

37 **[Intermediate]** • **Target: Geometry comparisons**

Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. Column B is greater

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Geometry comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater

39 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Column A: the value of x, if $x : 15 = 2 : 5$. Column B: 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. Column B is greater

40 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 3.01 Algebra Essentials

Two numbers are in the ratio $4 : 7$, and their sum is 99. Column A: the smaller number. Column B: 40.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. Column B is greater

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41 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 4.01 Geometry & Measurement

Column A: the sum of the parts in the simplified form of the ratio $24 : 36$. Column B: 5.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. Column A is greater

d. The two columns are equal

42 [Challenging] • Target: Ratio and proportion comparisons • Cross-Lesson: 6.01 Sequences & Patterns

A recipe uses water and rice in the ratio $2 : 1$. Column A: the water needed for 300 g of rice. Column B: 550 g.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information

43 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 7.01 Data Analysis & Probability

A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. Column A is greater

d. The relationship cannot be determined from the given information

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44 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 8.01 Word Problems & Reasoning

Column A: the range of the data set $6, 11, 19, 25$. Column B: 19.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. Column A is greater

45 [Challenging] • Target: Data and probability comparisons • Cross-Lesson: 1.01 Arithmetic & Number Sense

A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column A is greater

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